SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	03 JULY 2026
Team ID	
Project Title	**A Comprehensive Measure of Well-Being**
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to simplify the assessment of human development using machine learning. The system predicts the Human Development Index (HDI) score based on four key indicators: Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. The solution provides fast, consistent, and data-driven predictions through an easy-to-use Flask web application.

Project Overview	
Objective	The primary objective is to develop a machine learning-based system that predicts a country's Human Development Index (HDI) score using health, education, and income indicators.
Scope	The project covers data collection, exploratory data analysis, preprocessing, Linear Regression model development, evaluation, model saving, and integration with a Flask web application for real-time prediction.
Problem Statement	
Description	Manually analyzing multiple development indicators to estimate HDI is time-consuming and complex. Users need a simple system that can process key indicators and provide a quick and consistent HDI prediction.
Impact	The solution enables faster human development assessment, supports country-level analysis, helps identify development gaps, and provides useful insights for researchers, students, policymakers, and development organizations.
Proposed Solution	
Approach	Use machine learning to analyze the relationship between Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita, and the HDI score. A trained Linear Regression model generates predictions from user inputs through a Flask web application.
Key Features	• Machine learning-based HDI score prediction. • Uses four development indicators. • Interactive Flask web interface. • Exploratory data analysis and visualizations

	• Fast prediction without manual calculation. • Saved model for reuse without retraining.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications.	Standard multi-core CPU; dedicated GPU not required
Memory	RAM specifications	8 GB or above
Storage	Space for dataset, model, plots, and application	Minimum 1 GB free storage
Software		
Frameworks	Python frameworks	Flask, Scikit-learn
Libraries	Additional libraries	Pandas, NumPy, Matplotlib, Seaborn, Pickle/Joblib
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	HDI dataset; CSV format